



Power BI Data Analytics Assessment

Submission Report



GitHub Repository Link : [GitHub Repository](#)

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Tool	Microsoft Power BI
Dataset	FoodMart (1997–1998)

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1. Executive Summary

The FoodMart Power BI solution was developed to provide management with a consolidated view of sales, profitability, customers, products, returns and market performance for 1997–1998. The solution combines transaction files through the Folder Connector, relates them to customer, product, store, region and calendar lookup tables, and uses DAX measures to support management analysis.

The completed model records total revenue of \$1,764,546.44, gross profit of \$1,052,818.78 and a gross margin of 59.67%. A total of 833,489 units were sold to 8,842 distinct purchasing customers, while 8,289 units were returned, producing an overall return rate of approximately 0.99%.

Key KPI	Result
Total Revenue	\$1,764,546.44
Total Product Cost	\$711,727.66
Gross Profit	\$1,052,818.78
Gross Margin	59.67%
Total Quantity Sold	833,489
Distinct Purchasing Customers	8,842
Average Revenue per Customer	\$199.56
Total Returned Quantity	8,289
Return Rate	0.99%
High Priority Revenue	\$141,625.42
USA Fixed Revenue	\$1,177,956.44

2. Dataset Overview

The supplied data represents FoodMart retail operations and contains customer, product, store, region, calendar, transaction and return information.

Dataset	Rows	Purpose
Customer_Lookup	10,281	Customer demographics, membership and household attributes
Product_Lookup	1,560	Product, brand, retail price, cost, weight and product attributes
Store_Lookup	24	Store type, location, country and region identifiers
Region_Lookup	109	Sales district and sales region mapping
Calendar_Lookup	730	Date dimension covering the analysis period
Transactions 1997	86,837	1997 sales transactions
Transactions 1998	182,883	1998 sales transactions
Returns 1997–1998	7,087	Returned product quantities

Revenue is derived from transaction quantity multiplied by product retail price. Product cost is derived from transaction quantity multiplied by product cost. This enables profitability analysis even though revenue and cost are not stored directly in the transaction files.

3. Task 1 – Data Loading and Preparation

3.1 Data Loading

- Loaded all relevant FoodMart lookup datasets into Power BI.
- Loaded the 1997 and 1998 transaction files using the Power BI Folder Connector and combined them into one transaction fact table.
- Loaded the returns dataset for return analysis.

3.2 Data Preparation and Cleaning

- Verified column names and assigned appropriate data types for identifiers, dates, quantities, prices and descriptive fields.
- Prepared transaction and return dates for time-based analysis.
- Validated product, customer, store and region keys used to connect fact and dimension tables.
- Created calendar attributes such as Year, Month Name and Year-Month for trend analysis and slicers.
- Sorted Month Name by month number so January–December appear in chronological order.

3.3 Data Quality Considerations

The customer lookup contains 10,281 customer records, while the sales fact table contains 8,842 distinct customers with transactions. Therefore, the Total Customers KPI is intentionally based on distinct customer IDs in the transaction fact table; it represents purchasing customers rather than every customer stored in the lookup table.

4. Task 2 – Data Modelling

4.1 Star-Schema Design

The model follows a star-schema approach, with Fact_Transactions as the primary sales fact table and Fact_Returns as a secondary fact table. Lookup tables provide descriptive attributes used for filtering and analysis.

Table	Role	Typical Relationship
Fact_Transactions	Sales fact	Many-to-one to Customer, Product, Store and Calendar
Fact_Returns	Returns fact	Many-to-one to Product, Store and Calendar
Dim_Customer	Dimension	customer_id
Dim_Product	Dimension	product_id
Dim_Store	Dimension	store_id
Dim_Region	Dimension	region_id through store/region mapping
Dim_Calendar	Dimension	date to transaction/return date

4.2 Core DAX Measures

```
Total Revenue =  
SUMX(  
    Fact_Transactions,  
    Fact_Transactions[quantity] * RELATED(Dim_Product[product_retail_price])  
)  
  
Total Product Cost =  
SUMX(  
    Fact_Transactions,  
    Fact_Transactions[quantity] * RELATED(Dim_Product[product_cost])  
)  
  
Gross Profit =  
[Total Revenue] - [Total Product Cost]  
  
Gross Margin % =  
DIVIDE([Gross Profit], [Total Revenue])  
  
Total Quantity Sold =  
SUM(Fact_Transactions[quantity])  
  
Total Customers =  
DISTINCTCOUNT(Fact_Transactions[customer_id])  
  
Avg Revenue per Customer =  
DIVIDE([Total Revenue], [Total Customers])  
  
Total Returned Quantity =  
SUM(Fact_Returns[quantity])  
  
Return Rate % =  
DIVIDE([Total Returned Quantity], [Total Quantity Sold])
```

4.3 Time Intelligence

```
Revenue Previous Year =  
CALCULATE(  
    [Total Revenue],  
    SAMEPERIODLASTYEAR(Dim_Calendar[date])  
)  
  
YoY Revenue Growth =  
[Total Revenue] - [Revenue Previous Year]  
  
YoY Revenue Growth % =  
DIVIDE([YoY Revenue Growth], [Revenue Previous Year])
```

Revenue increased from \$565,238.13 in 1997 to \$1,199,308.31 in 1998. This is an increase of \$634,070.18, equivalent to approximately 112.18% year-over-year growth.

5. Required DAX Questions

5.1 Customer Priority Classification

Customers are classified as High Priority only when all three specified conditions are met: more than two children, Golden membership and homeowner status.

```
Customer Priority =  
IF(  
    Dim_Customer[total_children] > 2 &&  
    Dim_Customer[member_card] = "Golden" &&  
    Dim_Customer[homeowner] = "Y",  
    "High Priority",  
    "Normal"  
)
```

Using this rule, the lookup contains 725 High Priority customers; 620 of them appear in the transaction data.

5.2 Revenue from High Priority Customers

```
High Priority Revenue =  
CALCULATE(  
    [Total Revenue],  
    Dim_Customer[Customer Priority] = "High Priority"  
)
```

Result: \$141,625.42, representing approximately 8.03% of total revenue.

5.3 Fixed USA Revenue

```
USA Fixed Revenue =  
CALCULATE(  
    [Total Revenue],  
    FILTER(  
        ALL(Dim_Store[store_country]),  
        Dim_Store[store_country] = "USA"  
    )  
)
```

Result: \$1,177,956.44. The ALL function removes the current country filter before the USA condition is applied, so the USA revenue remains fixed even when Canada or Mexico is selected.

6. Task 3 – Dashboard Development

The dashboard was designed as a management-level reporting solution with separate pages for executive sales, product performance and customer analysis. Year, month, country and other relevant slicers allow management to interact with the report.

6.1 Executive Sales Overview

- KPI cards: Total Revenue, Total Customers, Gross Profit and Gross Margin %.
- Revenue by Store Type and Revenue by Sales Region.
- Revenue trend using a Year → Month hierarchy.
- Revenue by Country.
- Interactive slicers for Year, Month, Country, Region and Store Type.

6.2 Product Performance Analysis

- KPI cards: Total Revenue, Total Returned Quantity and Total Quantity Sold.
- Top product brands by revenue.
- Top products by gross profit.
- Top returned products.
- Bottom 5 products by revenue.
- Product-brand revenue distribution and interactive Product Brand filter.

6.3 Customer Analysis

- KPI cards: Total Customers, Average Revenue per Customer and High Priority Revenue.
- Revenue by customer occupation.
- Revenue by membership card.
- Revenue by yearly income segment.
- Revenue by Customer Priority.
- Customer distribution by gender.

6.4 Interactive Filtering and Drill-Down

The dashboard includes Year, Month, Country, Product Brand, Customer Priority, Region and Store Type filters where appropriate. The revenue trend uses a Year → Month hierarchy, allowing users to drill from annual performance into monthly detail.

6.5 Product Category Limitation

The supplied Product_Lookup table does not contain a product category, department, family or class field. Its available analytical attributes are product brand, product name, retail price, cost, weight, recyclable and low-fat indicators. Therefore, a true Product Category Performance analysis cannot be produced without inventing unsupported categories. Product Brand and Product Name were used as the closest valid product segmentation, and this limitation should be disclosed rather than fabricating a category field.

7. Task 4 – Business Insights and Recommendations

7.1 Five Key Business Insights

1. Strong year-over-year revenue growth

Evidence: Revenue increased from \$565,238.13 in 1997 to \$1,199,308.31 in 1998, an increase of \$634,070.18 or approximately 112.18%.

Classification: Trend / Opportunity

Management meaning: The business experienced substantial sales expansion in 1998. Management should identify which stores, regions and products drove this increase and determine whether the growth can be sustained.

2. Revenue is highly concentrated in the USA

Evidence: USA revenue is \$1,177,956.44, approximately 66.8% of total revenue. Mexico contributes \$478,915.66 and Canada contributes \$107,674.34.

Classification: Risk / Opportunity

Management meaning: The USA is the core market, but the high concentration creates dependence on one country. Mexico and Canada provide diversification opportunities.

3. North West is the strongest sales region

Evidence: The North West region generated approximately \$847,826.72, well ahead of Mexico Central (\$330,362.03) and South West (\$320,804.78).

Classification: Opportunity / Risk

Management meaning: North West practices may provide a model for other regions, while weaker regions such as Central West require investigation.

4. Customer revenue is concentrated in particular segments

Evidence: Bronze members generated approximately \$986,502.65, the largest membership contribution. Professional customers generated about \$573,998.70, and the \$30K–\$50K income segment generated about \$580,785.96.

Classification: Opportunity

Management meaning: These segments represent strong targets for retention, loyalty and personalized promotions.

5. Returns are low overall but specific products stand out

Evidence: The overall return rate is approximately 0.99%, but Hermanos Red Pepper recorded 17 returned units, while Hermanos Oranges and Walrus Merlot Wine each recorded 15.

Classification: Risk

Management meaning: Although the overall rate is low, recurring returns for specific products can indicate product-quality, expectation, packaging or operational issues and should be monitored.

7.2 Additional Product Findings

Finding	Result
Highest-revenue brand	Hermanos – \$56,659.48
Second-highest brand	Tell Tale – \$51,561.22
Highest-revenue product	Hermanos Green Pepper – \$2,489.70
Lowest-revenue product	Choice Bubble Gum – \$209.88
Highest-returned product	Hermanos Red Pepper – 17 units

The Bottom 5 revenue products are Choice Bubble Gum (\$209.88), Bird Call Silky Smooth Hair Conditioner (\$227.24), Top Measure Chardonnay Wine (\$240.12), Super Columbian Coffee (\$245.00) and Fast Buttered Popcorn (\$251.94). These products should be reviewed in context before discontinuation because low revenue can also result from limited availability or a narrow customer segment.

7.3 Three Actionable Recommendations

1. Diversify market growth beyond the USA

Maintain investment in the USA while developing targeted sales campaigns and store-level initiatives in Mexico and Canada. Track country revenue contribution over time to reduce excessive dependence on one market.

2. Use customer segmentation for targeted retention and upselling

Prioritize high-value segments such as Bronze members, Professional customers and the \$30K–\$50K income group. Use tailored promotions and membership-upgrade campaigns, while separately nurturing the High Priority segment that currently contributes about 8.03% of revenue.

3. Establish product performance and return monitoring

Create a recurring review of low-revenue and high-return products. Investigate pricing, product quality, placement, stock availability and customer feedback before making assortment decisions. Use return rate together with revenue and gross profit rather than returns alone.

8. Conclusion

The completed FoodMart Power BI solution transforms the supplied operational data into a structured management reporting model. The dashboard provides visibility into revenue growth, profitability, customer behavior, product performance, returns and regional performance. The analysis shows strong 1998 growth and a highly profitable sales base, but also highlights market concentration in the USA and product-level return issues that should be monitored. The recommended actions focus on market diversification, targeted customer strategies and disciplined product-performance management.



Executive Sales Overview

Country

- ☐ Canada
☐ Mexico
☐ USA

Region

All

Store Type

All

Year

- ☐ 1997
☐ 1998

Total Revenue



\$1.76M

Total Customers



9K

Gross Profit



\$1.05M

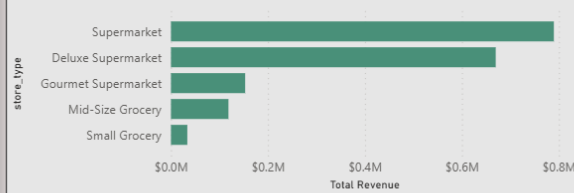
Gross Margin %

59.67%

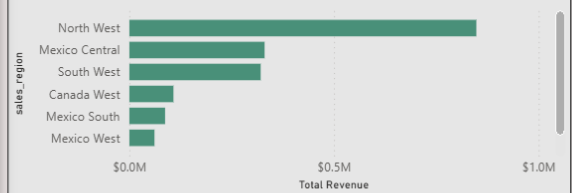
Month

- ☐ January
☐ February
☐ March
☐ April
☐ May
☐ June
☐ July
☐ August
☐ September
☐ October
☐ November
☐ December

Total Revenue by Store Type



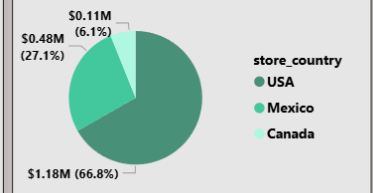
Total Revenue by Region



Total Revenue by Year & Month



Total Revenue by Country



Product Performance Analysis

Product Brand

All

Country

- ☐ Canada
☐ Mexico
☐ USA

Year

- ☐ 1997
☐ 1998

Total Revenue



\$1.76M

Total Returned Quantity

8K

Total Quantity Sold

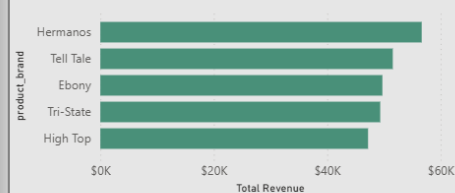


833K

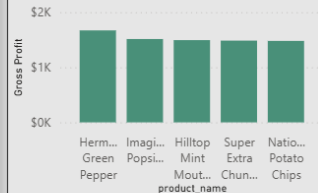
Month

- ☐ January
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☐ December

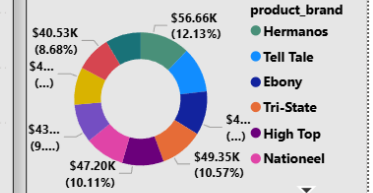
Total Revenue by Product Brand



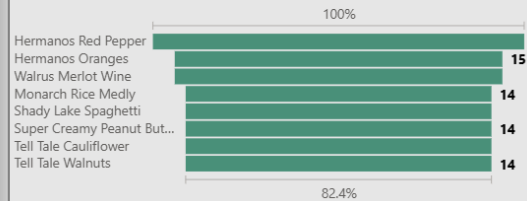
Gross Profit by Product Name



Total Revenue by Product Brand



Total Returned Quantity by Product Name



Bottom 5 Products by Revenue

